DeepCASE Project

Data input and graph construction

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Outline

Cross-host

- Generic input
- Historical host graphs
- LANL example

Input data

Each row describes an observed interaction between two computers:

$$e_i = (t_i, s_i, d_i, c_i).$$

Required field	Meaning
`timestamp`	t_i: event time in seconds
`source_computer`	s_i: computer initiating the event
`destination_computer`	d_i: destination computer
`event`	c_i: categorical description of the event

Optional fields include a unique `event_id`, `source_user`, `success`, and an evaluation `label`.

Historical graph snapshots

For events at time t, collect earlier activity in a window of width W:

$$\mathcal{H}_t = \{e_i : t - W \leq t_i < t\}, \qquad G_t = (V_t, E_t).$$

- **Nodes:** computers in the historical window, plus the source and destination computers of current events.
- **Directed edges:** one edge (s_i, d_i) for each distinct source–destination pair in $\mathcal{H}_t$.
- **Repeated interactions:** one edge per pair, with an interaction count and historical attributes.

All events at t share the same snapshot. Their categories, edges, and statistics enter history only at later timestamps.

Host and interaction features

For each computer

- Last L outgoing event categories, in chronological order.
- Counts of outgoing events, distinct source users, successes, and failures.

For each directed pair

- Interaction count and distinct source-user count.
- Failure fraction and age of the latest interaction.

Fit the category vocabulary on training data only. Reserve token 0 for padding and token 1 for unseen categories.

Counts and pair attributes use the whole window, even when a host's sequence exceeds L events.

LANL authentication input

LANL authentication records contain nine fields. Computers and accounts use consistent anonymous identifiers, and time counts seconds from the beginning of collection.

Illustrative authentication record

Field	Value
Time	`120`
Source user	`U7@DOM1`
Destination user	`U7@DOM1`
Source computer	`C10`
Destination computer	`C20`

Field	Value
Authentication	`Kerberos`
Logon type	`Network`
Orientation	`LogOn`
Result	`Success`

Source: LANL dataset documentation. This implementation reads `auth.txt.gz` for graph construction.

LANL normalization

The adapter preserves the timestamp and both host endpoints. It builds an event category from:

$$c_i = (\text{authentication type},\ \text{logon type},\ \text{orientation},\ \text{result},\ \text{user mismatch}).$$

In the worked example, abbreviate two categories as:

$$S = (\text{Kerberos}, \text{Network}, \text{LogOn}, \text{Success}, \text{False}),$$
$$F = (\text{NTLM}, \text{Network}, \text{LogOn}, \text{Fail}, \text{False}).$$

- Default scope: `LogOn` with distinct, known source and destination computers.
- Keep all authentication mechanisms, account categories, and success/failure outcomes within that scope.

LANL window example

Illustrative settings: $t = 200$ s and $W = 100$ s. The graph uses the interval $[100, 200)$.

All rows use `U7@DOM1` as both user accounts.

Time	Source	Destination	Category	Role at $t = 200$
90	`C30`	`C10`	S	Outside window
120	`C10`	`C20`	S	History
160	`C10`	`C20`	F	History
180	`C20`	`C30`	S	History
200	`C30`	`C10`	F	Current target

- The three historical rows create two directed edges.
- The two `C10` to `C20` interactions share one edge.
- The current event does not create a `C30` to `C10` edge in this snapshot.

LANL graph and host histories

The historical graph at $t = 200$ has three computer nodes:

$$\mathrm{C10} \xrightarrow{\text{2 events}} \mathrm{C20} \xrightarrow{\text{1 event}} \mathrm{C30}.$$

With $L = 3$, the outgoing host sequences and counts are:

Host	Sequence	Counts (n, u, s, f)
C10	$[S, F, 0]$	$(2, 1, 1, 1)$
C20	$[S, 0, 0]$	$(1, 1, 1, 0)$
C30	$[0, 0, 0]$	$(0, 0, 0, 0)$

S and F stand for category tokens, and 0 denotes padding. Counts represent outgoing events, distinct source users, successes, and failures. Node features apply $\log(1 + \text{count})$.

The target pair $(\mathrm{C30}, \mathrm{C10})$ has no edge in the window, so its historical pair attributes are zero.

Backup: feature definitions and evaluation labels

For a host v, let n_v, u_v, s_v, f_v count its outgoing events, distinct source users, successes, and failures:

$$x_v = [\log(1 + n_v), \log(1 + u_v), \log(1 + s_v), \log(1 + f_v)].$$

For an ordered pair (a, b), let n_{ab} count events, u_{ab} count distinct source users, and ρ_{ab} be its failure fraction:

$$a_{ab} = \left[\log(1 + n_{ab}), \log(1 + u_{ab}), \rho_{ab}, \frac{t - t_{ab}^{\text{last}}}{W}\right].$$

- GraphSAGE uses topology. The event head uses the target pair's historical attributes.
- Optional Red Team labels match (time, source user, source host, destination host) exactly and serve evaluation only.
- Label 0 means no known Red Team match.